

Program : Diploma in Communication and Computer Networking	
Course Code : 5327	Course Title: Fundamentals of Embedded System Design Lab
Semester : 5	Credits: 1.5
Course Category: Programme core course	
Periods per week: 3	Periods per semester: 45

Course Objectives:

- Implement simple programs using microprocessor.
- Implement interfacing with Arudino
- Implement interface with UART

Course Prerequisites:

Topic	Course code	Course Name	Semester
Concepts of 8051 microcontroller	4321	Introduction to Microcontroller	4

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	To understand the components of 8051	5	Applying
CO2	To understand the interfacing of 8051	10	Applying
CO3	To practice the interfacing of Arudino	11	Applying
CO4	To demonstrate the interfacing of UART	10	Applying
	Series Test	6	
	Open Ended Experiments / micro projects	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2				2		
CO3	2	2					
CO4	2		2				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	To write simple programming experiments with 8051		
M1.01	Familiarisation of 8051	6	Applying
M1.02	Addition of 8 bit numbers using 8051	3	Applying
M1.03	Subtraction of 8 bit numbers using 8051	3	Applying
CO2	To understand the interfacing of 8051		
M2.01	Using interfacing blink GREEN LED and RED LED	6	Applying
M2.02	Using interfacing provide delay and Blink GREEN LED and RED LED	6	Applying
	Series Test– I	2	
CO3	To practice the interfacing of 8051		
M3.01	Using Arudino intetrfacing Blink LED's	6	Applying
M3.02	Interfacing with 4 X 4 keyboard	3	Applying
CO4	To demonstrate interfacing of UART		
M4.01	Display a message using UART interface	3	Applying

M4.03	Develop a program to interface 7 segment Display with 8051	4	Applying
	Series Test -II	3	

Note: Experiments shall be conducted such that all COs are attained.

Minimum of 6 experiments (excluding open ended experiments/microprojects) shall be performed.

Compulsory for CIA

The CIA shall be arranged in third semester by the faculty in charge. The ESE need to be conducted at the end of the third semester.

Text / Reference

T/R	Book Title/Author
T1	Shibu.K.V, “Introduction to Embedded Systems”, TataMcgraw Hill,2009
T2	Peckol, “Embedded system Design”,JohnWiley&Sons,2010
T3	Lyla B.Das : Microprocessors and Microcontrollers, Pearson Education, India, 2011
R1	Rajkamal, ‘Embedded system-Architecture, Programming, Design’, TMH,2011
R2	Jorgen Staunstrup, Wayne Wolf ,Hardware / Software Co- Design Principles and Practice, Springer, 2009.
R3	Sam Siewert, V, Real-Time Embedded Components and Systems: With Linux and RTOS (Engineering), 2015
R4	Tanenbaum, Modern Operating Systems, 3/e, Pearson Edition, 2007.
R5	Wayne Wolf, Computers as Components: Principles of Embedded Computing System Design, 2/e, Kindle Publishers, 2005.

Online Resources

Sl.No	Website Link
1	https://onlinecourses.nptel.ac.in/noc21_ee18
2	https://www.udemy.com/course/mastering-microcontroller-with-peripheral-driver-development